

TECHNICAL MEMO

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To: StreamlineDB

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Date: January 22, 2026

Subject: NoSQL Architecture Recommendation for MVP

StreamlineDB's MVP is centered around user-generated content, dynamic user interaction and real-time engagement. Also, with the expectation of rapid evolution after launch, the database choice must prioritize flexibility, availability and horizontal scalability over a rigid structure. Based on these requirements, a document-oriented NoSQL database would fit best for MVP (Sullivan, 2015).

Database Type Selection

The recommended database for the primary data is a Document Database. StreamlineDB's data is inherently document-shaped, with the users' profiles varying significantly between creators and viewers, content types all carrying different metadata and the engagement data will be changing constantly. A document model allows these differences to exist naturally without forcing everything into a fixed schema (Sullivan, 2015). This is paramount for MVP. Features will be changing, new metadata fields will appear and requirements will shift once real users are onboarded. Also, document models allow related data to be stored together, reducing the need for expensive joins and improving the read performance for common access patterns like loading content alongside metadata and the engagement summaries.

CAP Theorem Analysis

For this environment the system should prioritize Availability and Partition Tolerance over having strict global consistency. In a distributed environment, network partitions are unavoidable and prioritizing availability ensures the platform remains usable during traffic spikes or partial failures (Sullivan, 2015). However, the trade-off will be that some data may be temporarily inconsistent across replicas. In practice, this will be acceptable for most user-facing

features. For example, the view count being slightly delayed or a like total updating a few seconds later does not meaningfully impact the user experience. StreamlineDB must ensure that the user can always browse content and interact in real time (Sullivan, 2015).

ACID vs BASE Decision

The database should primarily follow a BASE consistency model, with selective use of strong consistency where it is required.

Strong consistency is required for:

- User authentication and session handling
- Account creation and identify validation
- Permission checks

Eventual Consistency is acceptable for:

- View and like counts
- Follower totals
- Recommendation updates
- Analytics and engagement metrics
- Notification delivery

This approach allows the system to be able to remain fast and scalable without compromising the security or user trust. From an application design perspective this would mean validation and critical logic must be handled at the application layer rather than relying on rigid database constraints. Engagement data should be treated as aggregated or streaming data rather than single authoritative values (Sullivan, 2015).

Scaling Strategy

Document databases are built for horizontal scaling through sharding, which is where the data is distributed across multiple nodes based on a shared key. This will allow StreamlineDB to scale by adding servers instead of vertically upgrading hardware, which aligns with the unpredictable growth patterns that is expected with content platforms (Sullivan, 2015). The primary risk will come from uneven access patterns. Viral content or high-profile creators will introduce hot spots. But these can be mitigated by using hashed shard keys, separating write-heavy metrics from core content documents and then enforcing retention limits on high-volume data such as live chat logs. Also, analytics can be processed asynchronously to reduce load on primary collections (Sullivan, 2015).

Trade-off Analysis

Aspect	Document NoSQL Database	Relational Database
Schema Flexibility	Highly flexible; fields evolve naturally	Rigid schemas require migrations
Query Complexity	Optimized for content centric access patterns	Strong support for complex joins
Scaling Approach	Horizontal scaling via sharding	Primarily vertical scaling
Consistency Model	Tunable, eventual consistency where it is acceptable	Strong consistency by default
Development Speed	Faster MVP iteration	Slower due to schema constraints

For StreamlineDB's MVP, a document-oriented NoSQL database provides the perfect balance of flexibility, scalability and performance to meet its needs. It will be able to support rapid feature iteration, handle variable data cleanly and align with the availability requirements for a real-time content platform. While some consistency guarantees are eventual, those are done intentionally and are appropriate for early-stage growth (Sullivan, 2015).

References

Sullivan, D. (2015). *NoSQL for mere mortals*. Addison-Wesley.